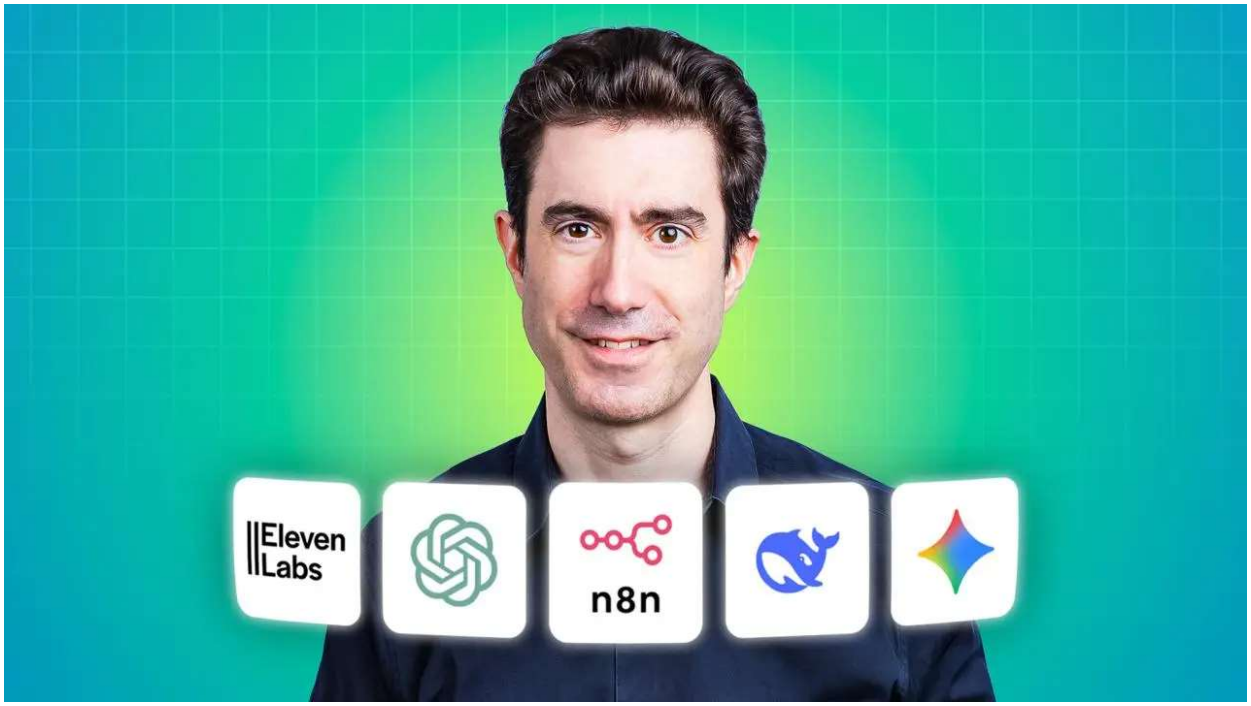


AI Builder with n8n – Create Agents and Voice Agents – RESOURCES

Jan 4, 2026 LLM Agent, AI, AI Engineering, ElevenLabs, n8n



Out of all the courses I've made, this has to be the single most satisfying and gratifying. It actually feels like it's CHEATING to create agents with n8n and ElevenLabs. Making substantial business impact simply shouldn't be this easy.

Well, I'm here to show you that it IS this easy!

After three short but intensive weeks, you'll be quite the expert. You'll deploy agents that rebalance your equity portfolio, answer questions with a RAG knowledge base, and drive sales to your business all the way through to booking demos.

You'll be positioned to build autonomous agents for yourself, your business and your clients that deliver significant value... in an insignificant amount of time.

It feels so productive and empowering to use n8n to build business solutions, that it feels more like entertainment than work – as you will see!

The Links

- The slides for the course, along with the Google Sheets and some json exports
- The Proficient AI Engineer directory of course graduates
- My digital twin that can answer questions on the course and curriculum

How much do I need to spend on n8n and APIs?

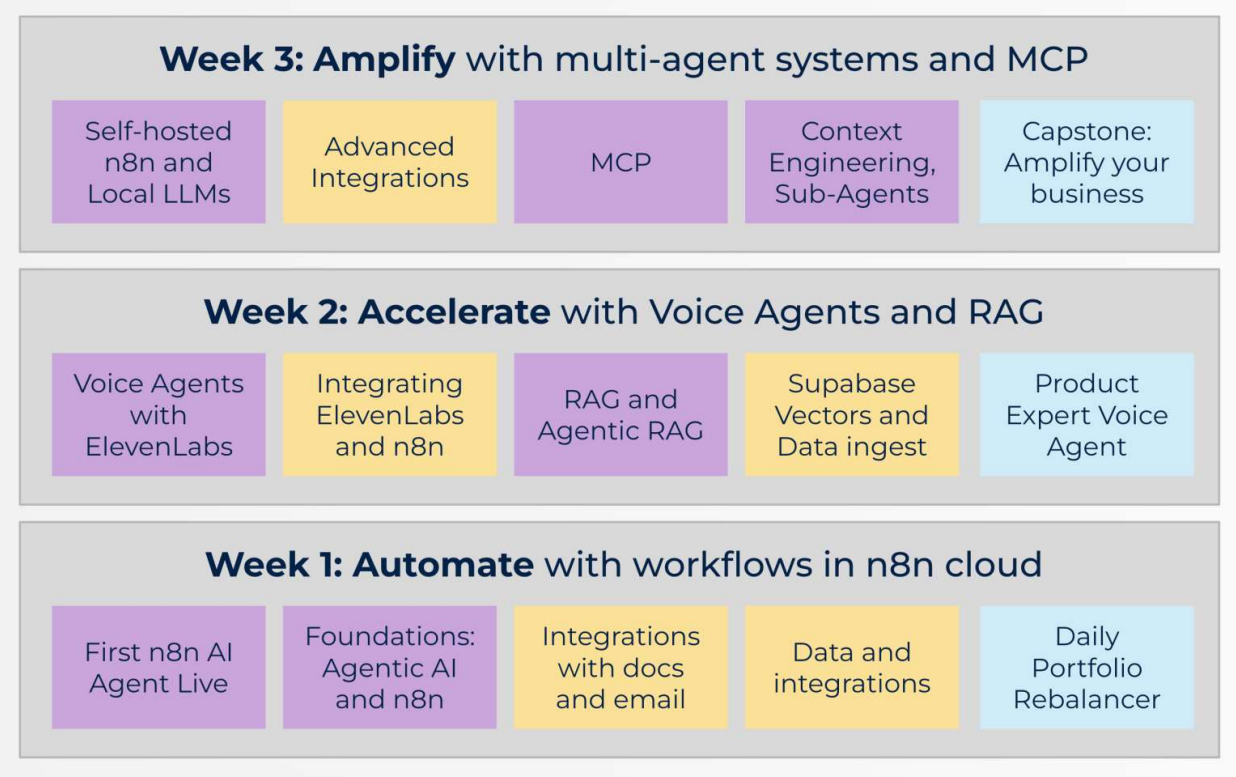
Absolutely nothing – everything we use on this course is free or optional. There are opportunities to spend if you have budget, but they are all optional.

There are 2 ways to run n8n:
n8n cloud: running on their infrastructure. This has a free trial period for at least 2 weeks, then a paid plan.
n8n self-hosted: free, running on your computer.

We start the course using n8n cloud to keep it simple, while you're on the free trial. Then I show you how to use the free n8n self-hosted in week 3, if you wish.

We work with many APIs on this course, including different LLMs. They are all optional. Free models via OpenRouter can be used throughout, but please see the note about privacy settings in the Week 1 Day 1 section below. ElevenLabs has a free trial and it's optional to use ElevenLabs.

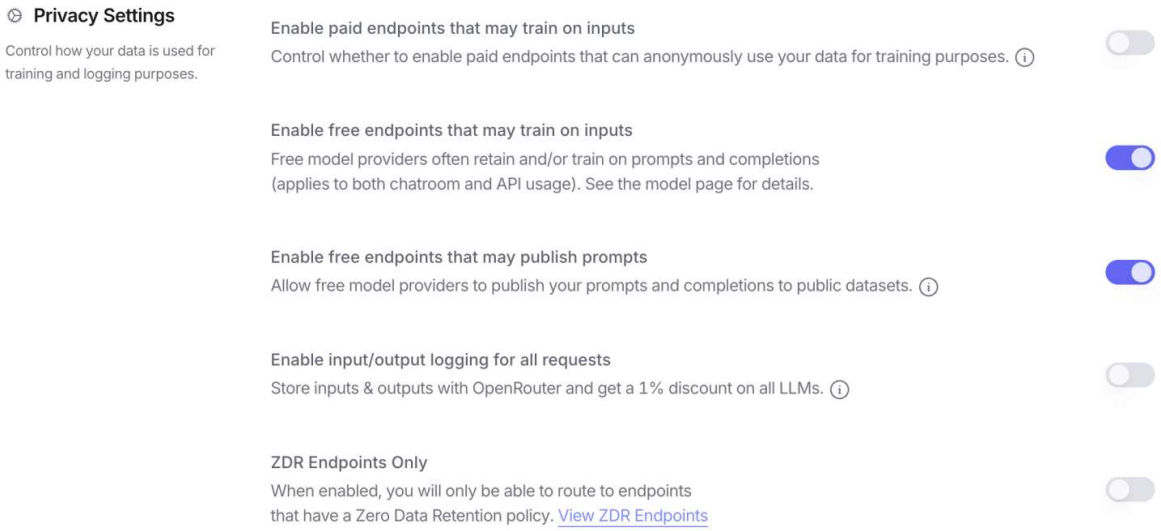
The Resources



Resources for Week 1

Link to n8n cloud: https://n8n.io

Week 1 Day 1:
If you wish to use free models, then when you set up OpenRouter, you'll need to go to this page:
https://openrouter.ai/settings/privacy
And set the following 2 fields to ON:
Enable free endpoints that may train on inputs
Enable free endpoints that may publish prompts



A pro-tip to keep in mind: if you get rate limits using free models on OpenRouter, you can pick the model **`"openrouter/free"`** as described here to be automatically routed to a working free model. Also see Q45 in the FAQ here.

If you'd like more background on what LLMs actually are and how they work, please check out my set of 4 videos here.:





Week 1 Day 3:

If the gmail tool doesn't look like mine and mentions "Human Review", you may have selected the wrong tool. Please see Q71 here and let me know if that doesn't fix it.

Week 1 Day 4:

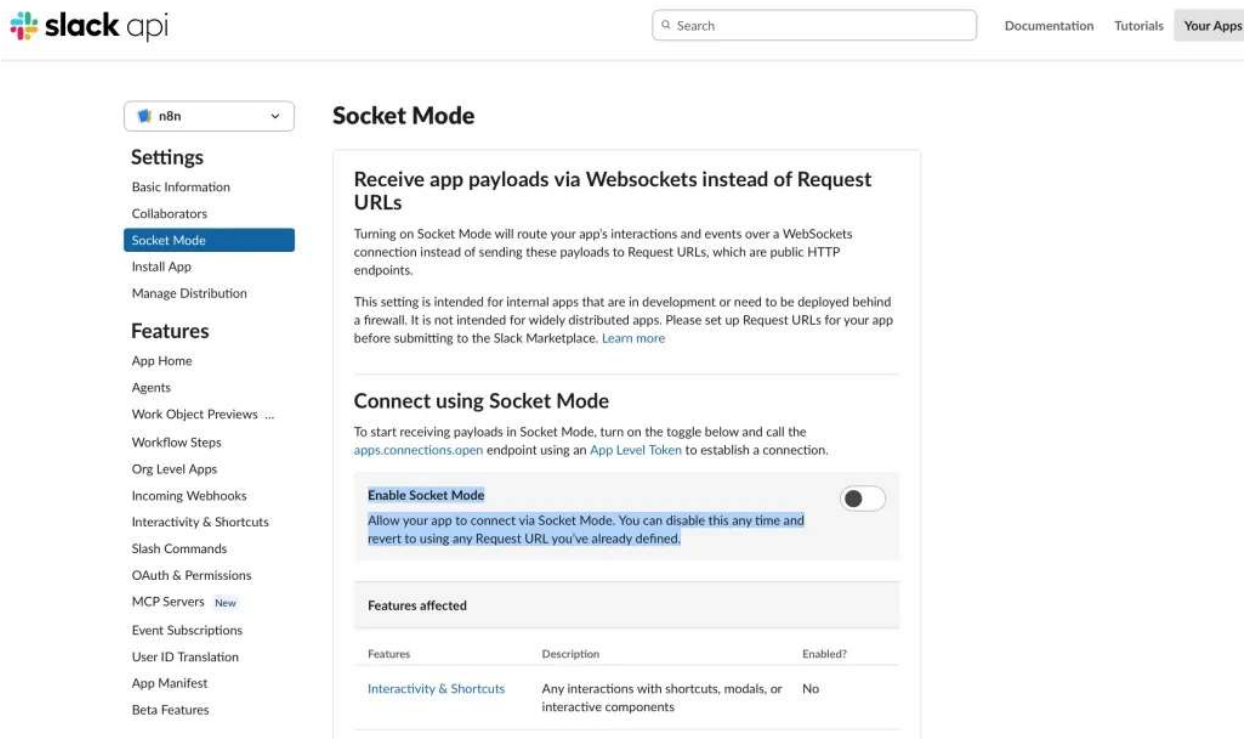
For Pushover push notifications, you may need to set the priority of the push in n8n to something higher than 'low' if you want your phone to make a sound, otherwise the push may arrive silently.

For Slack integration, please be sure not to set a 'Signing Secret' even though Slack recommends it. It adds complexity to the Slack integration and seems to make it sometimes not work. If you do want to use the Signing Secret, then here is a recommendation from student Emir (thank you!):

> In Slack, I went to Basic Information > Signing Secret > Regenerate > Copy. Then, inside the n8n workspace, in the Credentials section, I clicked the pencil/edit icon next to the credential and replaced the Signature

> Secret. After that, I clicked Execute step again, entered the URL inside the Enable Events field, and the verification passed successfully.

And also for Slack integration – make sure that "Connect using Socket mode" is DISABLED in the screen, as shown below. This may be the problem if your URL Request test works (verified), but your Slack Trigger doesn't work when you execute the workflow and mention your bot. Huge thanks to Trevor L for this, and Anjel L for confirming the fix.



Week 1 Day 5:

If you'd rather not create the workflow from scratch, I've put an export of the full workflow in the Week 1 folder here; after you create the Workflow, click on the three-dots menu in the top right and select "Import from File". You'll need to update the credentials and email address.

If you'd like to create the full workflow yourself, here is the User Prompt that I use, so you can copy and paste:

```
You have access to the user's Equity portfolio which is in Google Sheet. You also have access

{{ $json["How would you like to rebalance your portfolio?"] }}

In order to achieve this:

1. Read the portfolio from the sheet
2. Fetch latest prices of the positions
3. Update the table with the prices
4. Make decisions on portfolio rebalancing
5. Update the table with portfolio rebalancing decisions
6. Read the table again to check the rebalancing achieved the objective
7. Make any further changes to improve rebalancing, updating the table
8. Finally, send an email with trading decisions and concise push notification
```

And here is the prompt I use:

```
This tool provides details of the user's portfolio in Google Sheet. This includes the positio
```

Here is the portfolio sheet that I use – please save a copy of this in your Google Drive for your agent!

Resources for Week 2

Week 2 Day 2 – 4 notes on ElevenLabs, and the voice.html link:

- **Here is the voice.html page for submitting audio.**

4 notes on ElevenLabs: free plans, calling tools, navigation changes:

1. Some people on the free plan have mentioned that you need to select non-premium voices. “Bella” was mentioned as a voice that worked.
2. The free ElevenLabs plan may not support the TTS and STT API that we use at the start. If you’re on the free plan, please simply skip this first approach. The 2nd approach we use later in the day (with ElevenLabs calling n8n) is a far better approach, that is far more common in practice, and it should work fine with the free plan.
3. Agents in ElevenLabs sometimes don’t use Tools unless the Agent is published. If you’re having problems getting your Agent to call the n8n webhook, then please try publishing the Agent in ElevenLabs.
4. ElevenLabs recently rearranged their navigation menus. For example, Widgets is now under a ‘Channels’ section.

Week 2 Day 4: The Products sheet for the RAG information.

When you’re entering the Supabase details into n8n, there’s been a minor change in Supabase that you need to watch out for. The URL that displays in Supabase looks like this:
`https://xxxxx.supabase.co/rest/v1`
But the URL that you enter into n8n should not include the /rest/v1 suffix; so like this:
`https://xxxxx.supabase.co`

Here’s the SQL code to create the database table – the only code on the course! Notice the 1536 in 2 places here for the number of dimensions; you’ll need to change this if you’re using a different encoder.

```
create table if not exists knowledgebase (
  id uuid primary key default gen_random_uuid(),
  content text,
  metadata jsonb,
  embedding vector(1536)
);
create or replace function public.match_documents (
  query_embedding vector(1536),
  match_count int default 5,
  filter jsonb default '{}'
)
returns table (
  id uuid,
  content text,
```

```
  metadata jsonb,
  similarity float
)
language sql stable
as $$
  select
    id,
    content,
    metadata,
    1 - (embedding <=> query_embedding) as similarity
  from knowledgebase
  where metadata @> filter
  order by embedding <=> query_embedding
  limit match_count;
$$;
```

Resources for Week 3

Week 3 Day 1:

Link to Ollama: https://ollama.com

The docker command to run n8n self-hosted locally. (If you want schedulers inside n8n to have the right timezone, then replace `Europe/London` with your timezone from this document.). Please note that to stop n8n when it's running, simply press Ctrl+C in the Terminal / Powershell where it's running.

```
docker run -it --rm  --name n8n -p 5678:5678 -e GENERIC_TIMEZONE="Europe/London"  -e TZ="Euro
```

And later, the same docker command but with the ability to connect to the host so that Ollama can run locally:

```
docker run -it --rm --name n8n  -p 5678:5678 --add-host=host.docker.internal:host-gateway -e
```

Week 3, Day 2:
Here is a note from student Nguyen if you use self-hosted n8n, confirmed by another student (rephrased):

> I used n8n locally and followed the instructions for the HTTP node – Post, send Body, JSON, {{$json.data}} – but webhook.site returned blank in request content. I found a solution to use Raw in the body content type, application/json in the Content type, `{{ JSON.stringify($json.data) }}` in the body field.

The Google Drive OAuth process has been greatly simplified for n8n cloud – you can use the same approach as we did for Google Sheets; just click to sign in with your credentials. If you're using self-hosted n8n, you'll need to follow the same steps as I do on the video.

With thank you to student Omer C.: if you get this error during Google Drive OAuth2 API process:

> *n8n.cloud has not completed the Google verification process. The app is currently being tested, and can only be accessed by developer-approved testers. If you think you should have access, contact the developer.*

Then add your email address:

1. Go to https://console.cloud.google.com/auth/audience
2. Find "Test Users"
3. Click "+ Add user"

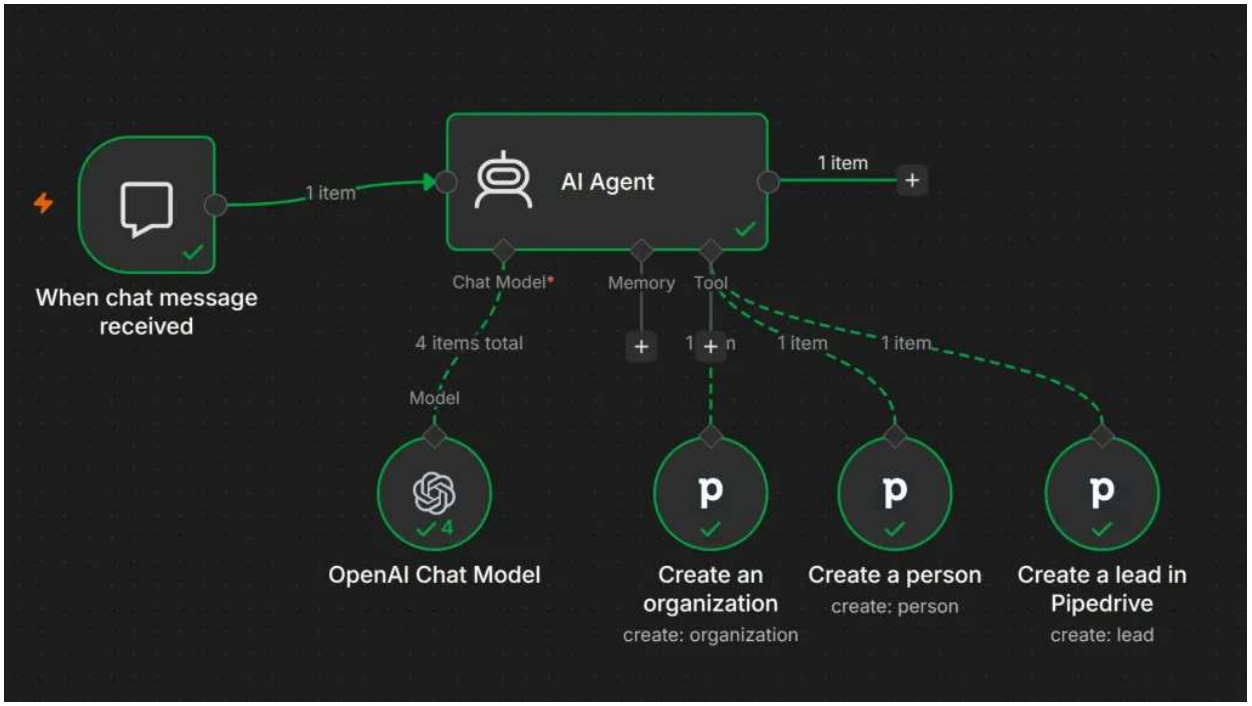
Week 3, Day 2/3 – FIRECRAWL

With thanks to Steve X: The Firecrawl node has been updated – the similar action I used is now under 'MAP & SEARCH ACTIONS' / 'Search the web and scrape results' instead of 'Search and optionally scrape search'. UPDATE: this has changed again (thank you Kadir S.) – in latest n8n this may be just "Search" and it combines web search with immediate, full-page content scraping.

If Project Variables aren't supported by your version of n8n, simply use the Firecrawl token directly in the URL. No need to upgrade to the paid version.

Week 3, Day 4: PIPEDRIVE

A student has noticed that Pipedrive now has simple Tools that make this far easier! Many thanks MJB for this; below is a screenshot of how this looks if you take advantage of the Tools:



Week 3, Days 4 and 5: Links, updates and prompts

The post on Context Engineering by Phil Schmid
And the seminal The Lethal Trifecta by the always brilliant Simon Willison.The ICP.pdf file to be dropped into your shared folder is in the week 3 folder here.

TWO NEW ISSUES RAISED IN JUNE 2026:

1. Some students have mentioned that there's a new issue in the Pipedrive integration: the AI Agent outputs a Person ID as a string rather a numeric value, and there's an error after it passes through IF node. The fix is to convert the Person ID into numeric value by replacing the $json value with:

`{{ parseInt($("AI Agent").item.json.output.id) }}`
But note the comment above that there's a simpler approach for Pipedrive..

2. Several students mentioned that they hit an Out Of Memory error on Day 5. I've been able to work around it using the following approach: I opened the Prospecting Sub-Agent and:
(1) I deactivated the Hunter MCP. Let's have it only use Firecrawl.
(2) I changed the System Prompt to this:
You find Sales Prospects based on a specific query from the user. You use your tools to search the internet and scrape websites as necessary to find the best potential clients for the user, and also use your tools to find the most likely email address for the prospect, and respond with 2 sales prospects. Do not ask questions or clarifications. Answer only with the final answer after using your tools.
(3) I changed the model to gpt-5.4-mini
With that, it worked fine without the memory error, and a student confirmed it worked for him too.

Here's the prompt used in ElevenLabs:

```
You are an Account Executive for Accounting Software company AccountAI.
```

I've put a full export of all the Capstone Project workflows in the week 3 folder here. I recommend you create this Project from scratch, but if you'd prefer to use mine, you can import from these files.

You'll need to create each Workflow, then click in the three-dots menu on the top right, and select "Import from File". You'll need to update the credentials, the Pushover User token, and your name & Voice Agent phone number.

Keep in mind: the Capstone project is a great end-to-end example and could form the basis of a real-world commercial product. Before taking it further, though, you'd need to make the Pipedrive query more scalable by only working with a subset of people at a time.

Thank you for making it to the very end! Please connect with me on LinkedIn, and/or sign up to stay in touch below. I can't wait to hear what you make of the course.

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One response to “AI Builder with n8n – Create Agents and Voice Agents – RESOURCES”



atul
January 22, 2026 at 2:32 am

its awesome

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